

ROMA FIUMICINO, Italy

WMO: 162420

Lat: 41.800N

Lon: 12.239E

Elev: 10

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 31.8	(c) 33.9	(d) 18.0	(e) 13.6	(f) 39.1	(g) 21.5	(h) 16.2	(i) 39.3	(j) 29.0	(k) 53.7	(l) 25.6	(m) 51.9	(n) 6.4	(o) 40	(p) 0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.0	89.0	71.4	86.6	71.9	85.1	71.8	77.2	83.1	75.8	82.2	74.5	81.2	10.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	132.9	80.6	73.7	125.4	79.7	72.0	118.4	78.6	40.6	83.3	39.2	81.7	38.0	81.5	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	20.1	17.5	27.1	93.4	2.5	2.3	25.3	95.0	23.8	96.4	22.4	97.6	20.6	99.3
			24.5	78.8	2.6	2.0	22.6	80.2	21.1	81.4	19.7	82.5	17.8	83.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.8	47.5	48.0	51.8	56.5	63.7	70.7	75.3	76.0	69.8	63.5	55.9	49.4		
	DBStd	11.07	5.02	4.99	4.30	4.25	4.08	4.39	3.27	3.44	4.19	4.45	5.67	5.37		
	HDD50	320	108	88	30	3	0	0	0	0	0	0	15	76		
	HDD65	2610	542	476	408	255	74	6	0	0	9	80	276	484		
	CDD50	4246	31	32	87	199	424	621	783	807	594	419	192	57		
	CDD65	1061	0	0	0	1	34	177	318	342	152	34	3	0		
	CDH74	8822	0	0	0	10	219	1428	2992	3199	884	89	1	0		
	CDH80	2378	0	0	0	1	26	326	881	1026	116	2	0	0		
(14) Wind	WSAvg	7.5	7.9	8.1	8.1	7.6	7.1	7.3	7.4	7.2	7.2	6.9	7.8	7.8		
(15) Precipitation	PrecAvg	27.9	3.0	2.8	2.4	2.0	1.3	0.8	0.3	1.1	2.8	3.5	4.0	3.5		
	PrecMax	46.6	8.4	6.8	6.3	5.6	4.6	4.3	4.1	4.7	11.0	10.6	14.4	8.8		
	PrecMin	15.9	0.1	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2		
	PrecStd	7.0	2.0	1.9	1.6	1.4	1.1	0.9	0.7	1.3	2.5	2.8	2.9	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	63.5	68.4	75.0	83.0	89.2	90.6	93.0	86.0	78.9	71.9	64.6		
		MCWB	56.3	54.0	55.3	60.6	64.3	70.1	70.4	71.5	70.2	67.3	64.0	58.5		
	2%	DB	60.1	60.7	64.8	71.4	79.0	86.1	88.1	89.7	82.7	76.6	69.5	62.4		
		MCWB	54.9	53.7	54.9	58.8	64.5	70.2	71.3	72.1	70.0	67.0	62.9	57.6		
	5%	DB	58.5	58.9	62.6	68.2	76.1	84.0	86.3	87.6	80.9	74.7	67.6	60.7		
		MCWB	53.6	52.8	54.9	58.3	64.0	69.8	71.9	72.6	69.8	66.1	61.5	56.1		
	10%	DB	56.9	57.3	60.8	66.0	73.5	81.1	84.6	85.6	78.9	72.5	65.3	59.0		
		MCWB	52.1	51.8	54.0	57.7	63.2	69.2	72.0	72.9	69.0	64.9	59.7	54.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.0	57.1	59.1	62.7	69.1	75.8	78.6	79.2	76.1	71.4	66.3	60.2		
		MCDB	61.1	60.2	64.1	70.0	76.0	81.7	83.9	84.1	80.4	75.3	69.6	62.9		
	2%	WB	55.8	55.3	57.4	61.0	67.3	74.0	76.9	77.7	73.5	69.2	64.0	58.4		
		MCDB	59.2	58.9	61.8	67.9	74.1	80.8	82.9	83.5	79.1	73.4	68.2	61.5		
	5%	WB	54.3	54.0	56.2	59.8	65.9	72.4	75.5	76.3	72.0	67.5	62.1	56.8		
		MCDB	57.8	57.7	60.6	66.1	72.8	79.8	82.0	82.7	78.1	72.5	66.3	60.2		
	10%	WB	52.5	52.5	55.0	58.7	64.6	71.0	74.3	75.0	70.5	66.0	60.3	54.8		
		MCDB	56.2	56.5	59.5	64.4	71.6	78.5	81.1	81.8	76.9	71.4	64.5	58.4		
(35) Mean Daily Temperature Range	MDBR		15.3	16.1	16.7	17.3	17.9	18.2	17.9	18.0	16.9	16.6	14.9	14.9		
	5% DB	MCDBR	15.4	16.5	19.2	20.4	21.9	21.3	20.2	20.5	18.2	17.6	14.9	14.2		
		MCWBR	11.5	11.9	12.3	11.6	10.8	10.2	10.5	10.5	10.3	10.7	10.0	10.7		
	5% WB	MCDBR	13.4	14.1	16.8	17.8	18.9	18.3	17.7	17.9	16.1	15.6	13.2	12.1		
		MCWBR	11.0	11.1	11.8	11.5	11.0	10.3	10.6	10.6	10.2	10.6	9.9	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.345	0.366	0.402	0.432	0.438	0.430	0.424	0.424	0.416	0.402	0.378	0.346		
	taud		2.452	2.371	2.270	2.202	2.235	2.305	2.326	2.338	2.354	2.368	2.365	2.446		
	Ebn at Noon		252	264	266	266	267	269	269	266	259	247	235	240		
	Edh at Noon		25	31	38	44	43	41	40	38	35	31	27	23		
(44) All-Sky Solar Radiation	RadAvg		561	858	1256	1682	2055	2328	2372	2070	1520	1034	636	506		
	RadStd		54	81	140	111	166	98	103	103	75	87	56	52		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.68	N/A	N/A	+1.13	N/A	N/A	N/A	N/A	+220	+119
(47) Regional (16 neighbors)		+0.76	N/A	N/A	+0.95	N/A	N/A	N/A	-140	+243	+124

Nomenclature: See separate page